

# Introduction to Robotics

Anatomy, Drives, End-Effectors, Controllers, Sensors & Economics

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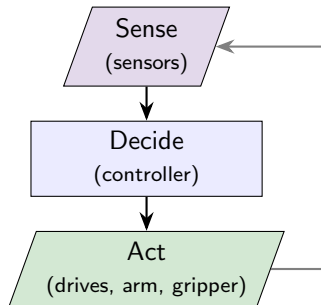
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# What is a Robot?

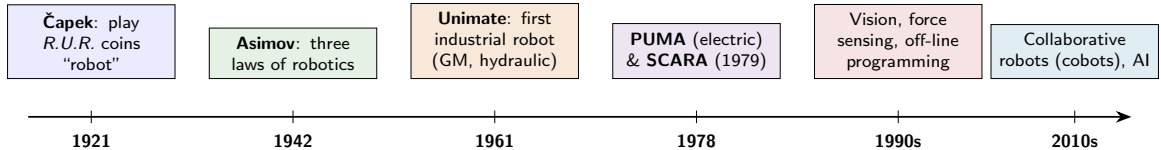
## Definition (Robot Institute of America)

An **industrial robot** is a **reprogrammable, multifunctional manipulator** designed to move material, parts, tools or special devices through **variable programmed motions** for the performance of a variety of tasks.

- **Reprogrammable** — a new task needs only a new program.
- **Multifunctional** — the same arm can weld, paint, load or assemble.
- **Manipulator** — an arm of links and joints that moves in space.
- A robot is the “general-purpose NC machine”: it follows the same **sense** → **decide** → **act** cycle.



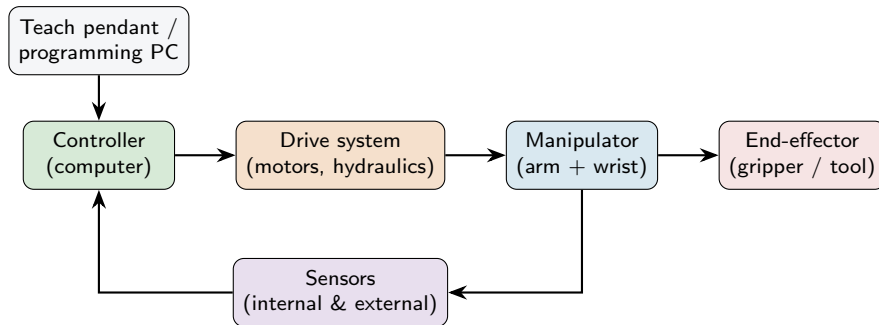
# A Short History of Robotics



## Asimov's three laws (1942)

A robot may not injure a human; it must obey orders unless that breaks the first law; it must protect itself unless that breaks the first two laws.

# Components of an Industrial Robot System



- **Manipulator** — the mechanical arm: base, links, joints and wrist.
- **Drive system** — powers the joints: hydraulic, pneumatic or electric.
- **End-effector** — the “hand” that does the job.
- **Controller** — stores the program, plans the motion, closes the servo loops.
- **Sensors** — tell the controller where the arm is and what is around it.

# Robot vs. Human Worker vs. NC Machine

| Feature         | Human worker     | Industrial robot             | NC machine           |
|-----------------|------------------|------------------------------|----------------------|
| Flexibility     | very high        | high (reprogram)             | medium (one process) |
| Repeatability   | poor, tires      | excellent ( $\pm 0.02$ mm)   | excellent            |
| Hazardous areas | unsafe           | safe                         | —                    |
| Judgement       | excellent        | limited (sensors, AI)        | none                 |
| Working hours   | 1 shift, breaks  | 24 $\times$ 7                | 24 $\times$ 7        |
| Typical job     | complex, one-off | repetitive handling, welding | cutting metal        |

## Key idea

Robots replace people in jobs that are **dangerous, dirty, dull** or need **high repeatability** — the “3-D” rule.

# Links, Joints and Kinematic Pairs

## Building blocks of a mechanism

A **link** is a rigid body; two links connected so that relative motion is **constrained** form a **kinematic pair (joint)**. Links + joints form a **kinematic chain**.

| Joint (pair)    | Allowed motion                   | DOF ( $f$ ) | Example           |
|-----------------|----------------------------------|-------------|-------------------|
| Revolute (R)    | rotation about one axis          | 1           | elbow, hinge      |
| Prismatic (P)   | sliding along one axis           | 1           | telescopic arm    |
| Screw (H)       | rotation coupled to sliding      | 1           | lead screw        |
| Cylindrical (C) | rotation + sliding, same axis    | 2           | piston in bore    |
| Spherical (S)   | rotation about 3 axes            | 3           | ball joint, wrist |
| Planar (E)      | 2 slides + 1 rotation in a plane | 3           | block on a table  |

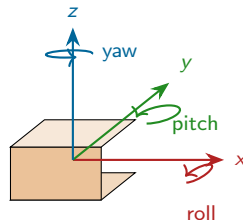
Industrial robots use almost only **R** and **P** joints (1 DOF each) — one actuator per joint.

# Degrees of Freedom and Movability Constraints

## Degrees of freedom (DOF)

A free rigid body in space has **6 DOF**: 3 translations ( $x, y, z$ ) + 3 rotations (roll, pitch, yaw). In a plane it has **3 DOF**.

- Every joint **removes** some freedoms — a revolute joint leaves only 1 of 6.
- To place *and* orient an object anywhere in space the robot needs **6 DOF**: 3 for the arm (position) + 3 for the wrist (orientation).
- Fewer DOF  $\Rightarrow$  restricted tasks; more than 6  $\Rightarrow$  **redundant** robot (can reach around obstacles).



3 translations + 3 rotations = 6 DOF

# Mobility (Grübler–Kutzbach) Criterion

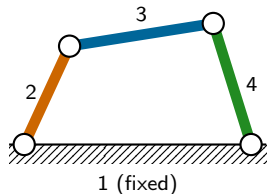
Mobility  $M$  = number of independent inputs needed

**Spatial:** 
$$M = 6(n - 1 - j) + \sum_{i=1}^j f_i$$

**Planar:** 
$$M = 3(n - 1) - 2j_1 - j_2$$

$n$  = links (*including* the fixed base),  $j$  = joints,  $f_i$  = DOF of joint  $i$ ,  $j_1$  = 1-DOF joints,  $j_2$  = 2-DOF joints.

- $M > 0$  — a **mechanism** with  $M$  inputs.
- $M = 0$  — a rigid **structure**.
- $M < 0$  — over-constrained (pre-loaded) structure.
- **Synthesis:** choose links & joints so that  $M$  equals the number of motors we want.



Four-bar:  $n = 4$ ,  $j_1 = 4$

$$M = 3(4 - 1) - 2(4) = \boxed{1} \text{ — one motor drives it.}$$

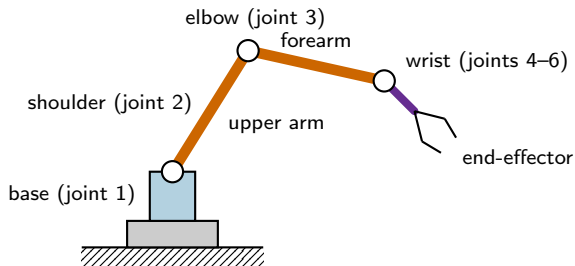
# Mobility: Worked Examples

| Chain                      | n | Joints | M | Working                     |
|----------------------------|---|--------|---|-----------------------------|
| Planar triangle (3 bars)   | 3 | 3 R    | 0 | $3(2) - 2(3) = 0$ structure |
| Planar four-bar            | 4 | 4 R    | 1 | $3(3) - 2(4) = 1$           |
| Planar five-bar            | 5 | 5 R    | 2 | $3(4) - 2(5) = 2$ 2 motors  |
| Planar 3R open arm         | 4 | 3 R    | 3 | $3(3) - 2(3) = 3$           |
| Spatial 6R robot arm       | 7 | 6 R    | 6 | $6(7 - 1 - 6) + 6 = 6$      |
| Spatial 7R (redundant) arm | 8 | 7 R    | 7 | $6(8 - 1 - 7) + 7 = 7$      |

## Serial vs. parallel robots

**Serial** (open chain): one joint after another — large workspace, errors add up. **Parallel** (closed chain, e.g. Stewart platform, delta robot): several legs share the load — stiff, fast, but small workspace.

# Robot Anatomy: Base, Arm, Wrist, End-Effector



- **Base** — fixed to the floor, ceiling or a moving track.
- **Arm (body + arm)** — first 3 joints; **position** the wrist in space.
- **Wrist** — last 2–3 joints (roll, pitch, yaw); **orient** the tool.
- **End-effector** — gripper or tool mounted on the wrist flange.
- **Tool centre point (TCP)** — the point the controller moves and programs.

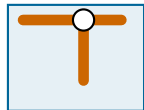
# Types of Robot Joints

| Symbol | Joint      | Relative motion                                             | Axes of input / output links |
|--------|------------|-------------------------------------------------------------|------------------------------|
| L      | Linear     | sliding, links <i>parallel</i>                              | same axis                    |
| O      | Orthogonal | sliding, output link <i>perpendicular</i>                   | at right angles              |
| R      | Rotational | rotation, axis <i>perpendicular</i> to both links           | e.g. elbow                   |
| T      | Twisting   | rotation, axis <i>parallel</i> to both links                | e.g. wrist roll              |
| V      | Revolving  | rotation, input link parallel, output perpendicular to axis | e.g. base swivel             |

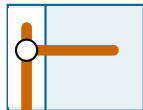
## Notation

A robot is described by its arm joints followed by its wrist joints, e.g. **TRR:RT** = articulated arm with a two-joint wrist. L and O are **prismatic**; R, T and V are **revolute**.

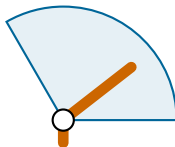
# The Five Common Robot Configurations



**Cartesian**  
LOO  
box envelope



**Cylindrical**  
TLO  
hollow cylinder



**Polar**  
TRL  
partial sphere



**SCARA**  
VRO  
annulus (top view)



**Articulated**  
TRR  
irregular

- **Work envelope** (work volume) — all points the wrist can reach; its shape follows from the joint types.
- **Articulated** (jointed-arm) is the most common; **SCARA** is the fastest for assembly; **Cartesian** (gantry) is the most accurate & easiest to program.

# Comparing Robot Configurations

| Configuration | Joints | Advantages                      | Typical applications                      |
|---------------|--------|---------------------------------|-------------------------------------------|
| Cartesian     | LOO    | rigid, accurate, simple control | gantry loading, 3-D printing, palletising |
| Cylindrical   | TLO    | simple, large vertical reach    | machine loading, assembly                 |
| Polar         | TRL    | long horizontal reach           | spot welding, die casting (Unimate)       |
| SCARA         | VRO    | very fast, stiff vertically     | electronic assembly, pick-and-place       |
| Articulated   | TRR    | dexterous, compact footprint    | arc welding, painting, general handling   |

## Wrist motions

**Roll** (twist about the arm axis, T), **pitch** (up-down, R) and **yaw** (left-right, R). A 3-DOF arm + 3-DOF wrist = full 6-DOF robot.

## Key specifications

- **Payload** — mass it can carry at full speed (kg).
- **Reach** — maximum distance from base to wrist.
- **Speed** — tool speed (m/s) or cycle time.
- **Spatial resolution** — smallest move it can make.
- **Accuracy** — how close it reaches a *commanded* point.
- **Repeatability** — how close it returns to a *taught* point ( $\pm 3\sigma$ ).

**Example:** a sliding joint has a range of 1.0 m and its position is stored as an **8-bit** number. Mechanical errors have  $\sigma = 0.5$  mm.

$$CR = \frac{1000}{2^8} = \frac{1000}{256} = \boxed{3.9 \text{ mm}}$$

$$\text{Accuracy} = \frac{CR}{2} + 3\sigma = 1.95 + 1.5 = \boxed{3.45 \text{ mm}}$$

$$\text{Repeatability} = \pm 3\sigma = \pm 1.5 \text{ mm}$$

With 12 bits,  $CR = 1000/4096 = 0.24$  mm — more bits, finer control.

# Forward Kinematics of a Two-Link Arm

## Forward kinematics

Given the **joint angles**, find the **position** of the gripper.

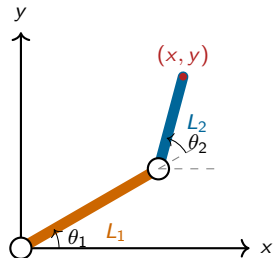
$$x = L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2)$$

$$y = L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2)$$

## Inverse kinematics

Given the **position**, find the **angles** — what the controller must solve to move in straight lines:

$$\cos \theta_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2}$$



$$L_1 = 0.5 \text{ m}, L_2 = 0.3 \text{ m}, \theta_1 = 30^\circ, \theta_2 = 45^\circ$$

## Pseudocode

- 1 START
- 2 Set  $L_1 = 0.5$ ,  $L_2 = 0.3$
- 3 READ th1, th2 (degrees)
- 4 Convert to radians
- 5  $x = L_1 \cos \text{th1} + L_2 \cos(\text{th1} + \text{th2})$
- 6  $y = L_1 \sin \text{th1} + L_2 \sin(\text{th1} + \text{th2})$
- 7 PRINT x, y
- 8 STOP

## C Program (forward kinematics)

```
#include <stdio.h>
#include <math.h>

#define PI 3.14159265

int main(void)
{
    float L1 = 0.5, L2 = 0.3;  /* metres */
    float t1, t2, x, y;

    printf("th1 th2 (deg): ");
    scanf("%f %f", &t1, &t2);

    t1 = t1 * PI / 180.0;      /* to rad */
    t2 = t2 * PI / 180.0;

    x = L1*cos(t1) + L2*cos(t1 + t2);
    y = L1*sin(t1) + L2*sin(t1 + t2);

    printf("x = %.3f m, y = %.3f m\n", x, y);
    return 0;
}
```

## Program Output

th1 th2 (deg): 30 45

x = 0.511 m, y = 0.540 m

## Check by hand:

$$x = 0.433 + 0.078 = 0.511$$

$$y = 0.250 + 0.290 = 0.540$$

Compile with gcc fk.c

-o fk -lm (-lm links the maths library for sin/cos).

## Hydraulic

### Pressurised oil

- Very high payload & power-to-weight.
- Stiff, smooth, holds heavy loads.
- Leaks, noise, maintenance.
- Early robots (Unimate), heavy handling, forging.

## Pneumatic

### Compressed air

- Cheap, fast, clean, spark-free.
- Air compresses → poor position control.
- Usually moves between **end stops**.
- Pick-and-place, small parts, grippers.

## Electrical

### Servo & stepper motors

- Accurate, quiet, clean, easy to control.
- Needs gearing (harmonic drive, belts).
- Lower power-to-weight than hydraulics.
- **Most modern robots.**

Choice depends on **payload, accuracy, speed, environment** and **cost**.

# Comparing Robot Drive Systems

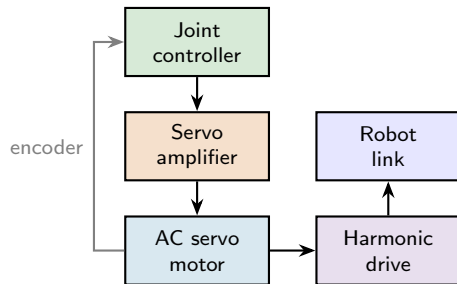
| Feature         | Hydraulic                | Pneumatic           | Electrical                  |
|-----------------|--------------------------|---------------------|-----------------------------|
| Payload         | very high (100s of kg)   | low (a few kg)      | low to high                 |
| Positioning     | good (servo valves)      | poor, end-stop only | excellent                   |
| Speed           | medium                   | very high           | high                        |
| Explosive areas | safe (no sparks)         | safe                | needs special motors        |
| Cleanliness     | oil leaks                | clean               | clean                       |
| Maintenance     | high                     | low                 | low                         |
| Typical robot   | heavy / outdoor handling | pick-and-place unit | welding, assembly, painting |

## Key idea

The same ideas you met for fluid-power and electrical control systems apply here — a robot is simply several such actuators arranged as a **kinematic chain**.

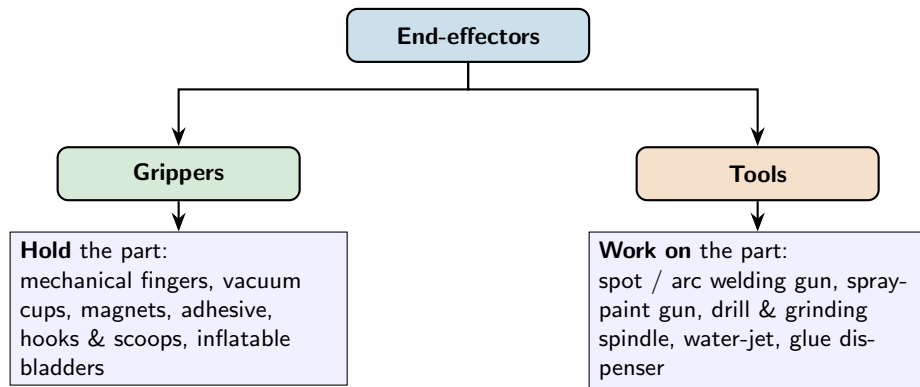
# Electrical Manipulators: Motors and Transmissions

- **AC servo motors** (brushless) with encoders — standard for industrial arms.
- **Stepper motors** — small, low-cost, open-loop arms and grippers.
- **Harmonic (strain-wave) drive** — ratios 50–160:1 in one stage, near-zero backlash.
- **Ball screws, belts, rack & pinion** — linear axes of gantry robots.
- **Brakes** hold the arm when power fails.



**Example:** motor 3000 rpm, harmonic drive 100:1  $\Rightarrow$  joint speed = 30 rpm =  $180^\circ/\text{s}$ , with  $100\times$  the motor torque.

# End-Effectors: Grippers and Tools



- The end-effector is usually **custom-designed** for the job — the robot arm itself is general-purpose.
- **Tool changers** let one robot pick up different end-effectors automatically.

# Mechanical Grippers

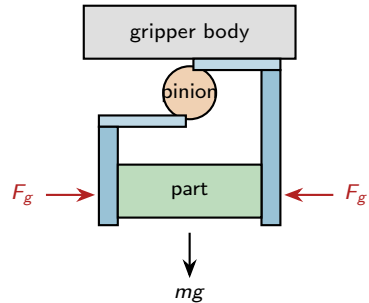
## Ways of holding

**Friction grip** — fingers squeeze the part; friction carries the weight.

**Physical constraint** — finger shape surrounds the part (form closure).

## Actuating mechanisms

- Linkage (toggle) mechanisms.
- Rack & pinion / gear drives.
- Cam and screw mechanisms.
- Pneumatic cylinder or electric motor as the input.



Rack-and-pinion parallel gripper: one input moves both fingers equally.

# Gripper Design: Required Gripping Force

## Friction grip

The friction force of all fingers must carry the weight *and* the inertia force of the part:

$$\mu n_f F_g \geq m(g + a) \implies F_g = \frac{m(g + a)}{\mu n_f} \times \text{SF}$$

$\mu$  = coefficient of friction,  $n_f$  = number of contact surfaces (fingers),  $a$  = acceleration, SF = safety factor.

**Problem:** a 2 kg part is held between two fingers ( $\mu = 0.3$ ). The robot accelerates it upwards at  $10 \text{ m/s}^2$ . Use  $\text{SF} = 2$ .

$$F_g = \frac{2(9.81 + 10)}{0.3 \times 2} = 66.0 \text{ N}$$

$$F_g (\text{SF} = 2) = \boxed{132 \text{ N}}$$

Rubber finger pads (higher  $\mu$ ) lower the force needed — and the risk of crushing the part.

# Vacuum, Magnetic and Other Grippers

## Vacuum cups

Lift force  $F = \Delta p A$ . Ideal for flat, smooth sheets — glass, sheet metal, cartons.

**Example:** lift a 5 kg steel plate with one cup, vacuum  $\Delta p = 60 \text{ kPa}$ ,  $SF = 2$ .

$$A = \frac{SF \cdot mg}{\Delta p} = \frac{2 \times 5 \times 9.81}{60\,000} = 1.64 \times 10^{-3} \text{ m}^2$$

$$D = \sqrt{\frac{4A}{\pi}} \approx \boxed{46 \text{ mm}}$$

## Magnetic grippers

Electromagnets or permanent magnets for **ferrous** parts — fast pick-up of steel plates, any shape; residual magnetism must be removed.

## Others

Adhesive grippers (fabric, light parts), hooks & scoops (bags, liquids), inflatable bladders (fragile or irregular parts), soft robotic fingers.

# End-Effector Design Considerations

| Consideration     | Question to ask                                                        |
|-------------------|------------------------------------------------------------------------|
| Part properties   | mass, size, shape, surface finish, fragility, temperature, magnetic?   |
| Gripping method   | friction or form closure? vacuum or magnet possible?                   |
| Force & weight    | grip force with SF; end-effector mass counts against the robot payload |
| Tolerances        | how accurately must the part be located? need compliance?              |
| Sensing           | part-present switch, force sensor, vision for location?                |
| Power & utilities | air, vacuum, electric — routed through the wrist                       |
| Changeover        | quick-change or automatic tool changer for different parts?            |

## Remote centre compliance (RCC)

A passive wrist device that lets the part **tilt and shift slightly** during insertion — so a peg can enter a hole even when the robot is a little misaligned.

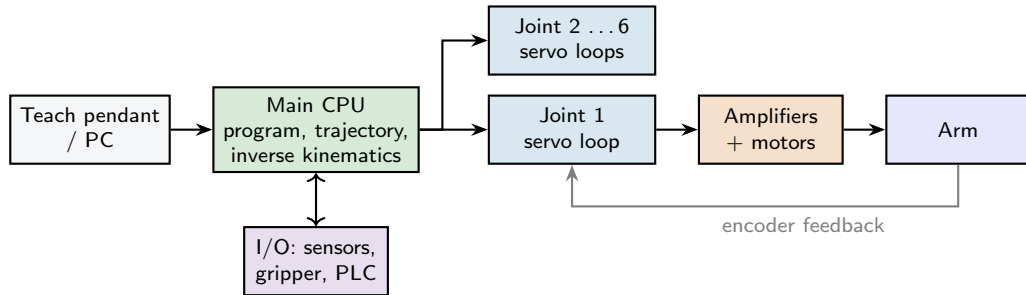
# Levels of Robot Control

| Type                         | How it works                                                                   | Controller               |
|------------------------------|--------------------------------------------------------------------------------|--------------------------|
| Limited-sequence (bang-bang) | joints move between <b>mechanical stops</b> ; sequence set by a timer or logic | fluidic logic, PLC, cams |
| Playback, point-to-point     | taught points stored; path between them not controlled                         | microprocessor           |
| Playback, continuous path    | whole path controlled (straight lines, arcs) by interpolation                  | microprocessor / DSP     |
| Intelligent                  | uses sensors & vision to change its actions; makes decisions                   | multi-processor, PC, AI  |

## Key idea

The simplest robots need only **ON/OFF logic** — which can be built from **air** (fluidics). Servo-controlled robots need a **microprocessor** to calculate and close the loops.

# Microprocessor-Based Robot Controller



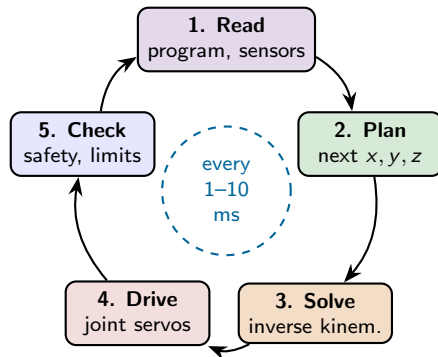
- **Main processor:** interprets the program, plans the path, solves inverse kinematics every few ms.
- **Joint processors / DSPs:** run a PID position loop for each joint at 1–10 kHz.
- **I/O:** signals to/from grippers, conveyors, machine tools and safety fences (interlocks).

# How a Robot Controller Runs: The Motion Cycle

## The Read—Plan—Solve—Drive cycle

Just like the CPU's machine cycle, the robot controller repeats one cycle every **interpolation period** (typically 1–10 ms) while the robot is moving.

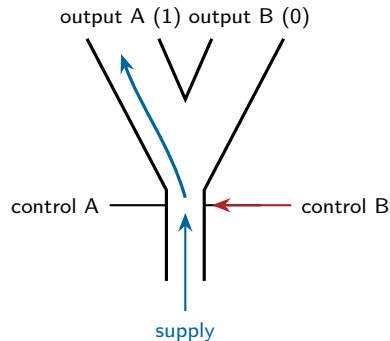
- 1 **Read** — next program step and all sensor inputs.
- 2 **Plan** — next point on the path (straight line / arc) in  $x, y, z$ .
- 3 **Solve** — inverse kinematics: joint angles for that point.
- 4 **Drive** — send the angles to the joint servo loops (PID).
- 5 **Check** — encoders, limits, collision & safety signals; then repeat.



## Fluidics

Logic built from **air (or liquid) jets** in channels with **no moving parts**. A jet sticks to one wall (**Coanda effect**); a small control jet switches it to the other wall — a binary 0/1 element.

- Gates: AND, OR, NOT, NOR, flip-flop (memory), timers.
- Drive pneumatic cylinders directly — ideal for **limited-sequence** pick-and-place robots.
- Immune to heat, radiation, magnetic fields, explosive atmospheres.
- Slow (ms) and bulky compared with electronics; no arithmetic.



Wall-attachment flip-flop: a pulse at *control B* pushes the jet to output B, where it stays.

# Microprocessor vs. Fluidic Control

| Feature       | Microprocessor                          | Fluidic                                    |
|---------------|-----------------------------------------|--------------------------------------------|
| Signal        | electrical (bits in memory)             | air-jet pressure (0/1)                     |
| Speed         | $\mu$ s per operation                   | ms per operation                           |
| Functions     | arithmetic, kinematics, PID, vision     | ON/OFF logic, sequencing only              |
| Programming   | software — easy to change               | re-plumb the circuit                       |
| Environment   | needs protection from heat, EMI, sparks | works in explosive, hot, radioactive areas |
| Typical robot | servo, continuous-path, intelligent     | limited-sequence, pneumatic                |

## Key idea

Today microprocessors dominate; fluidics survive where **electricity is dangerous** — e.g. paint booths and cargo areas with flammable vapour.

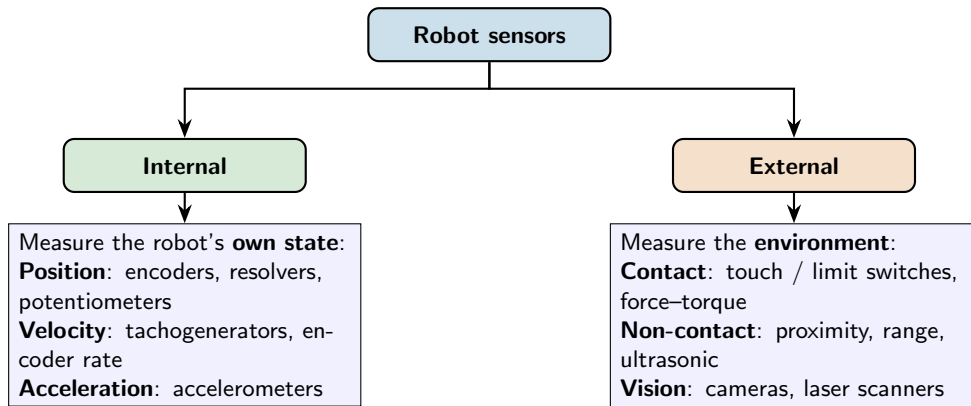
## Methods

- **Lead-through (manual)** — operator physically guides the arm; motion recorded (painting).
- **Teach pendant** — jog the robot with buttons, store points.
- **Robot language** — text program (VAL, KRL, RAPID).
- **Off-line** — simulate in software, download the program.

## A pick-and-place program (VAL style)

```
PROGRAM PICKPLACE
  SPEED 50           ; % of maximum
  OPENI             ; open gripper
  APPRO PICK, 50     ; 50 mm above
  MOVES PICK        ; straight down
  CLOSEI            ; grip the part
  DEPARTS 50         ; lift straight up
  APPRO PLACE, 50
  MOVES PLACE
  OPENI              ; release
  DEPARTS 50
END
```

# Classification of Robot Sensors



Internal sensors close the **servo loops**; external sensors let the robot **react** to the world.

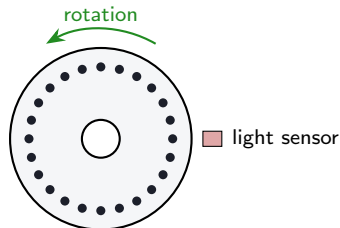
# Internal Sensors: Encoders

## Incremental encoder

A disc with  $N$  slots gives  $N$  pulses/rev; two channels (A, B)  $90^\circ$  apart give direction and  $4N$  counts/rev (quadrature). Needs a **homing** run at start-up.

## Absolute encoder

Each position has a unique binary (Gray) code on  $n$  tracks:  $2^n$  positions/rev. Knows its position at power-on.



### Examples:

12-bit absolute:  $360^\circ/4096 = 0.088^\circ$

1000-line incremental,  $\times 4$ :  $360^\circ/4000 = 0.09^\circ$

Through a 100:1 gear the joint resolution becomes 100 times finer.

# External Sensors: Touch, Force, Proximity and Range

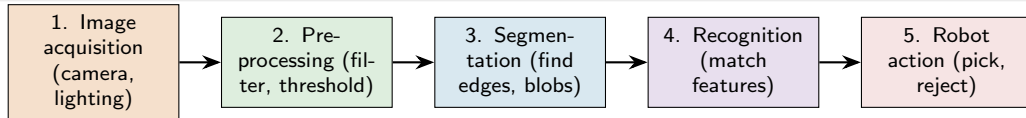
| Sensor               | Principle                             | Robot use                      |
|----------------------|---------------------------------------|--------------------------------|
| Touch / limit switch | contact closes a switch               | part present in gripper        |
| Tactile array        | grid of pressure cells in finger pads | shape & slip detection         |
| Force–torque (wrist) | strain gauges measure 6 components    | assembly, polishing, collision |
| Inductive proximity  | eddy currents from metal (a few mm)   | detect metal parts             |
| Capacitive proximity | capacitance change, any material      | level, non-metal parts         |
| Ultrasonic / optical | time of flight or reflected light     | distance, obstacle avoidance   |

## Range by time of flight

$$d = \frac{v t}{2} \quad \text{e.g. ultrasonic echo after } t = 2 \text{ ms, } v = 343 \text{ m/s} \Rightarrow d = \frac{343 \times 0.002}{2} = \boxed{0.343 \text{ m}}$$

## Machine vision

A camera and computer that let the robot **see**: find a part's position and orientation, check its quality, or read codes.



- **Bin picking** — find randomly placed parts; **seam tracking** — follow a weld joint.
- **Inspection** — measure dimensions, detect defects at production speed.
- Good **lighting** is half the job — back-lighting gives sharp silhouettes.

# Characteristics of Jobs Suitable for Robots

A job is a good robot candidate when it is

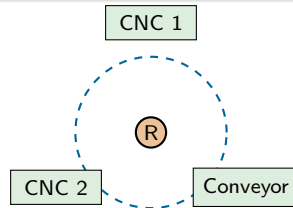
- **Hazardous** or uncomfortable for people (heat, fumes, paint, radiation).
- **Repetitive** with a steady work cycle.
- Handling parts that are **heavy** or awkward.
- Run over **multiple shifts**.
- Needs **consistent quality** (welds, paint thickness).
- Parts arrive at a **known position** and orientation.

Work-cell layouts

**Robot-centred** — robot in the middle serving machines around it.

**In-line** — robots along a conveyor (car body welding).

**Mobile** — robot on a track or AGV serving several machines.



# Applications of Industrial Robots

## Material handling

- Pick-and-place, transfer.
- Palletising / packing.
- **Machine loading:** CNC, presses, die casting, forging.

## Processing

- **Spot & arc welding.**
- **Spray painting**, coating.
- Grinding, deburring, polishing, cutting (laser, water-jet).

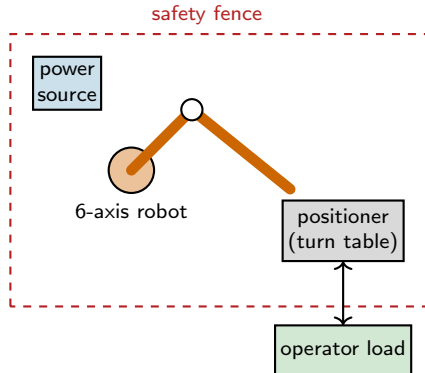
## Assembly & inspection

- Inserting, screwing, gluing.
- Electronic PCB assembly (SCARA).
- Vision / probe inspection.

## Marine and shipbuilding applications

Welding robots for hull blocks and double-bottom sections; plasma / laser profile cutting of plates; hull-blasting and painting crawlers; underwater ROVs for hull inspection, cleaning and offshore maintenance; tank-cleaning robots.

# Application Example: Robotic Arc Welding Cell



- Robot carries a MIG/MAG welding torch as its end-effector.
- **Positioner** turns the part so welds are made in the flat (down-hand) position.
- **Seam tracking** (arc or laser sensor) corrects for fit-up errors.
- Two stations: the operator loads one while the robot welds the other.
- Typical gain: arc-on time rises from  $\sim 30\%$  (manual) to 80–90 %.

# Costs and Benefits of a Robot Installation

## Investment costs

- Robot arm + controller.
- End-effectors, fixtures, positioners.
- Safety fencing, sensors, conveyors.
- Engineering, installation, programming, training.

The robot itself is often only **30–50 %** of the total project cost.

## Operating costs vs. savings

- **Costs:** maintenance, power, tooling, programming changes.
- **Savings:** direct labour, higher output, less scrap & rework, lower injury costs, consistent quality.

## Justification methods

Payback period, return on investment (ROI), net present value (NPV).

# Payback and ROI: Worked Example

## Relations

$$\text{Payback} = \frac{\text{total investment}}{\text{net annual saving}}$$

$$\text{ROI} = \frac{\text{net annual saving}}{\text{total investment}} \times 100 \%$$

**Problem:** a machine-loading robot costs Rs 30 lakh; installation, gripper and fencing add Rs 10 lakh. It replaces one operator on each of two shifts *and* a relief worker for each (4 people, Rs 4 lakh/year each). Robot running costs are Rs 4 lakh/year.

$$\text{Investment} = 30 + 10 = 40 \text{ lakh}$$

$$\text{Labour saving} = 4 \times 4 = 16 \text{ lakh/yr}$$

$$\text{Net saving} = 16 - 4 = 12 \text{ lakh/yr}$$

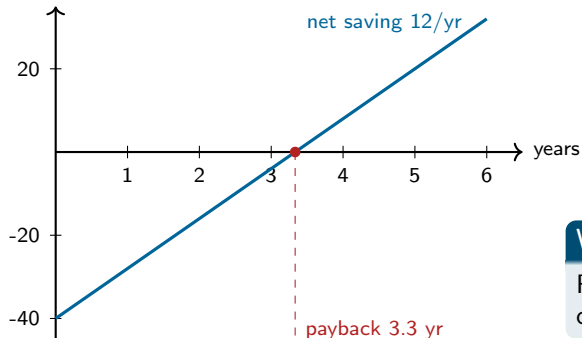
$$\text{Payback} = 40/12 \approx \boxed{3.3 \text{ years}}$$

$$\text{ROI} = 12/40 = \boxed{30 \%}$$

Most companies accept a robot project if the payback is below 2–3 years; here extra output or quality savings would be needed.

# Cumulative Cash Flow and Break-Even

cumulative cash (Rs lakh)



- Start at -Rs 40 lakh (investment).
- Each year adds Rs 12 lakh of net savings.
- The line crosses zero at the **payback point**.
- Over a 6-year life the robot earns Rs 32 lakh more than it cost.

## Ways to improve the case

Run 3 shifts, serve 2 machines with one robot, count quality & safety savings.

# Summary & What Comes Next

## Concept

- Robots: definition, history, system components.
- Links, joints, DOF and the mobility criterion.
- Robot anatomy, configurations, specifications, kinematics.
- Hydraulic, pneumatic and electrical manipulators.
- End-effectors and gripper design.
- Microprocessor and fluidic controllers; programming.
- Sensors, applications and economics.

## Practice makes perfect

- Compute the mobility of mechanisms around you.
- Run the kinematics C program for other angles; add inverse kinematics.
- Size a gripper for a part of your choice.

## Coming up

Robot kinematics & dynamics, trajectory planning, and marine / underwater robotics.

# Thank You

Questions?